Day 059 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Apply Expression Structure

Standards: 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4c, 6.EE.A.4

Learning target: I can apply Expression Structure, and explain and check my reasoning.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapters 5, 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Work inside parentheses, then evaluate exponents. Multiply and divide from left to right; then add and subtract from left to right. Multiplication does not automatically come before division. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units. Find the GCF of the two addends. Divide each addend by that factor and place the quotients inside parentheses. The inside numbers should share no common factor greater than 1. Use operation properties to rewrite an expression. Matching values at one input do not prove equivalence for all inputs. A single counterexample is enough to show two expressions are not equivalent.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 8, 14, 17, 18

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01 / Learn

Work inside parentheses, then evaluate exponents. Multiply and divide from left to right; then add and subtract from left to right. Multiplication does not automatically come before division. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units. Find the GCF of the two addends. Divide each addend by that factor and place the quotients inside parentheses. The inside numbers should share no common factor greater than 1. Use operation properties to rewrite an expression. Matching values at one input do not prove equivalence for all inputs. A single counterexample is enough to show two expressions are not equivalent.

Worked example 1

Evaluate 8 + 3 × 6. Show the intermediate steps.

1. Work inside parentheses, then evaluate exponents.

2. Multiply and divide from left to right; then add and subtract from left to right.

3. Use parentheses and exponents first, then × and ÷ left to right, then + and -.

4. Result: 26.

5. State the result in the form requested: 26.

Answer: 26

Worked example 2

Select all expressions equivalent to 7(x + 4).
A. 11x
B. 7x + 28
C. 7x + 4
D. 28 + 7x

1. Use operation properties to rewrite an expression.

2. Distribute 7 to both terms to get 7x + 28.

3. Reversing the addends gives the same value.

4. The other expressions fail, for example at x = 0.

Answer: B, D - 7x + 28; 28 + 7x

Day 059 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Evaluate 2 + 5 × 5. Show the intermediate steps.

02 / Constructed response

Evaluate 4x + 3 for x = 2.

03 / Constructed response

Factor 8 + 12 using its GCF. Make the inside numbers relatively prime.

04 / Constructed response

Are 7(x + 3) and 7x + 21 equivalent for every x? Use a property or a counterexample.

05 / Constructed response

Evaluate (8 + 6) × 3^2. Show the intermediate steps.

06 / Constructed response

Evaluate x^2 + 8y for x = 5 and y = 7.

07 / Constructed response

Factor 32 + 48 using its GCF. Make the inside numbers relatively prime.

08 / Multiple select

Select all expressions equivalent to 3(x + 2).
A. 6 + 3x
B. 5x
C. 3x + 6
D. 3x + 2

09 / Numeric entry and error analysis

Evaluate $4^2 + 2(8 - 3)$. Part A: enter the value. Part B: a student begins by adding 4^2 and 2; explain why that changes the expression.

10 / Constructed response

A rectangle has length x + 4 and width 7. Use A = lw to write its area expression and find A when x = 4.

Day 059 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Apply Expression Structure

Standards: 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4c, 6.EE.A.4

Learning target: I can apply Expression Structure, and explain and check my reasoning.

1. 27

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 27.

6.EE.A.1, 6.EE.A.2c | Constructed response

2. 11

Substitute first: 4 × 2 + 3 = 11.

6.EE.A.2c | Constructed response

3. 4(2 + 3)

GCF = 4; 8 = 4 × 2 and 12 = 4 × 3.

6.NS.B.4c | Constructed response

4. Yes; distribution gives 7x + 21

7(x + 3) = 7x + 21. This matches.

6.EE.A.4 | Constructed response

5. 126

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 126.

6.EE.A.1, 6.EE.A.2c | Constructed response

6. 81

Substitute first: 5^2 + 8 × 7 = 81.

6.EE.A.2c | Constructed response

7. 16(2 + 3)

GCF = 16; 32 = 16 × 2 and 48 = 16 × 3.

6.NS.B.4c | Constructed response

8. A, C - 6 + 3x; 3x + 6

Distribute 3 to both terms to get 3x + 6. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

9. 26

Parentheses: 8 - 3 = 5. Exponent: $4^2 = 16$. Multiply: 2 × 5 = 10. Add: 16 + 10 = 26. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

10. A = 7(x + 4); 56 square units

Substitute x = 4: 7(4 + 4) = 56.

6.EE.A.2c | Constructed response